

Distributed Clinical Platform Architecture & Data Flow Overview

System Topology • Zero-Trust Medical Telemetry • Document ID: MY-ARCH-2026-V2

1. Architectural Paradigm

Moyassar Health AI utilizes a microservices-based, zero-trust cloud architecture designed for high availability, sub-second inference latency, and strict healthcare compliance. The infrastructure decouples client interactions from specialized deep-learning workers and external large language model (LLM) reasoning cores.

2. Core Infrastructure Layers

Layer	Component Role	Protocols & Specifications	Security Standards
Ingress / Gateway	TLS Termination & Routing	TLS 1.3, HTTPS, Anycast CDN	Rate Limiting, WAF, CORS
Diagnostic Workers	Computer Vision & ML Inferences	Decoupled Microservice Containers	Ephemeral Memory, No PII
Reasoning Core	Anthropic Claude 3.7 / 3.5 Sonnet	Encrypted REST API Calls	Zero Data Training Policy
Client Access	Hospital, Doctor & Patient Portals	Role-Based Access Control (RBAC)	JWT Authentication, MFA

3. Secure Telemetry Pipeline

When an imaging scan or laboratory panel is submitted:

- **Client Encryption:** Imagery is encrypted via AES-256 in transit and assigned a stateless session identifier.
- **Isolated Inference:** Deep learning workers perform feature extraction and spatial bounding in sandboxed, non-persistent memory.
- **Claude Reasoning Pipeline:** Extracted clinical metrics are formatted into structured JSON tokens and dispatched securely to Anthropic's Claude API.
- **Dual Reporting:** Structured physician SOAP notes and patient-accessible guidance are returned and rendered.

4. Data Protection & Regulatory Compliance

Moyassar adheres strictly to HIPAA Security Rule 45 CFR Part 160 and Part 164. All external AI queries are governed under Anthropic's commercial privacy agreement, ensuring customer data is never used to train or fine-tune public models.